

EMCH 260

SOLID MECHANICS

Ming Hu

Associate Professor
Department of Mechanical Engineering
University of South Carolina
E-Mail: hu@sc.edu

Mechanical Properties of Materials

3

Chapter 3.4

- ✓ Understand more deeply the plastic and elastic deformation, Yield strength in idealized elastoplastic materials
- ✓ Know how to analyze plastic strain and permanent set in idealized elastoplastic materials

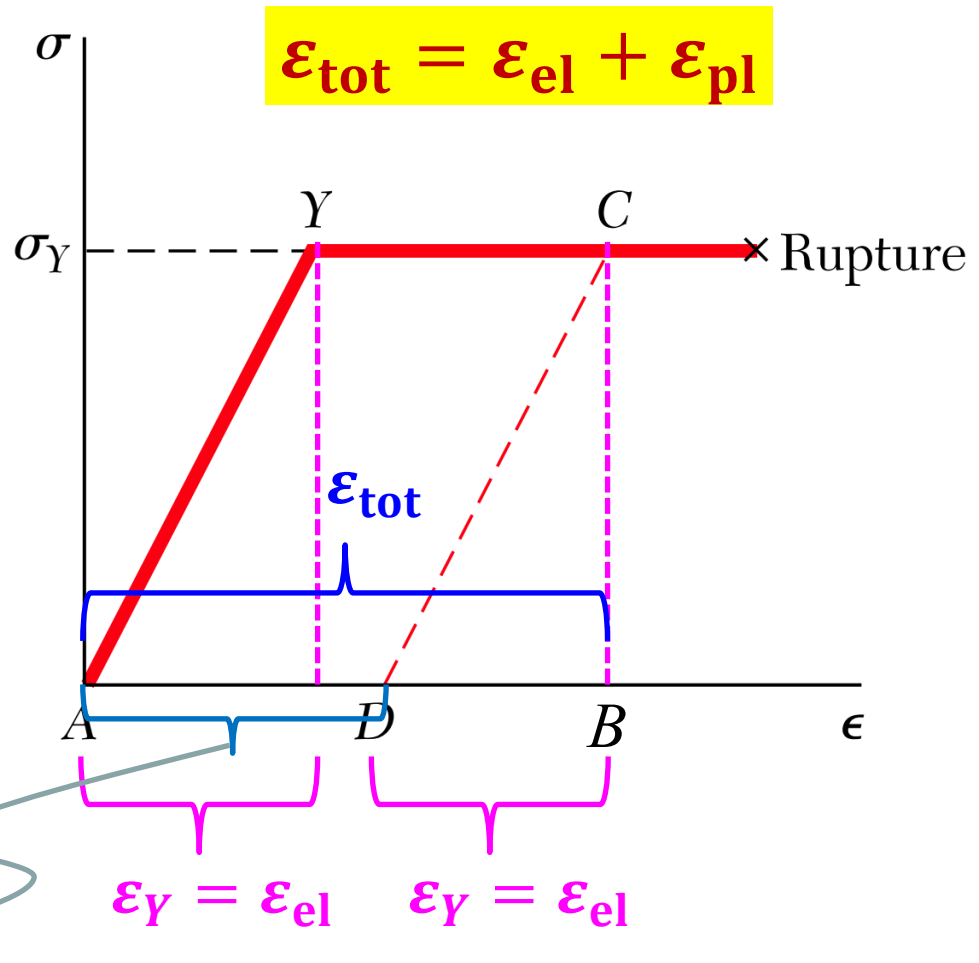
IDEALIZED ELASTOPLASTIC MATERIAL

- The stress-strain diagram consists of the two straight-line segments

- When σ reaches the value σ_Y , the material starts yielding and keeps deforming plastically under a constant load.

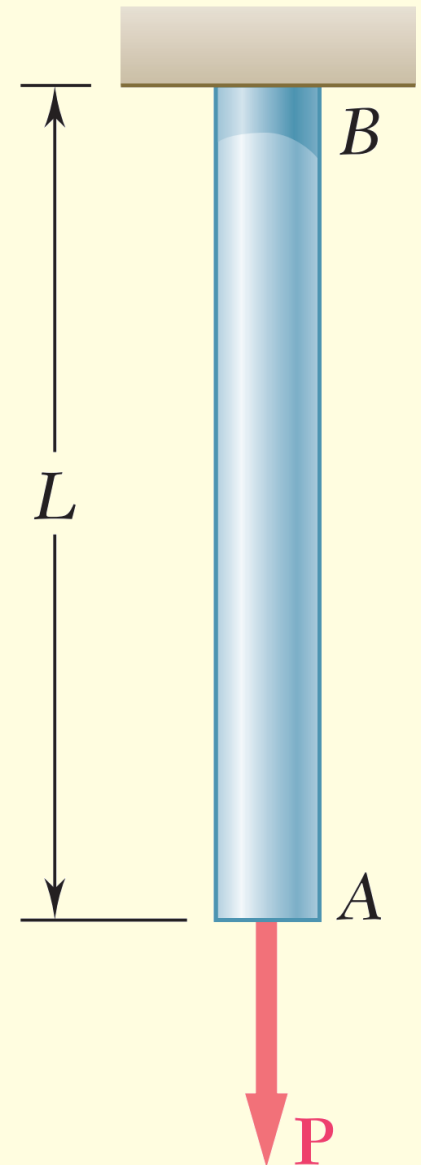
- If the load is removed, unloading takes place along a straight-line segment CD parallel to the initial portion AY of the loading curve.

- The total strain is equal to the recovered part (ϵ_{el}) plus the permanent set part (ϵ_{pl})



IN-CLASS EXAMPLE

The cylindrical rod AB has an initial length $L_0 = 1.8$ m and a diameter of $D = 3.2$ cm. It is made of a mild steel that is assumed to be elastoplastic with $E = 200$ GPa and $\sigma_Y = 250$ MPa. A force P is applied to the rod until the end A has moved down by an amount $\delta_m = 3.5$ mm. Determine the maximum value of the force P and the permanent set (δ_p) of the bar after the force has been removed.



IN-CLASS EXAMPLE (cont'd)

Solutions

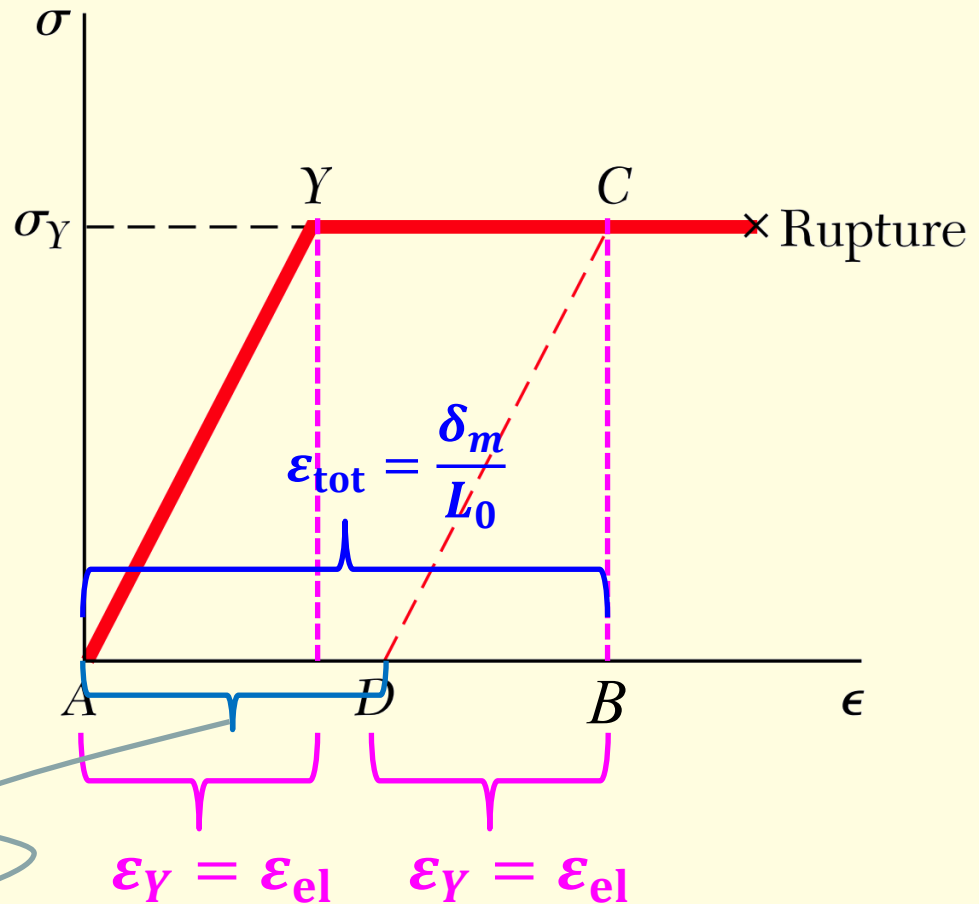
Total strain in the rod before removing load

$$\begin{aligned}\epsilon_{\text{tot}} &= \frac{\delta_m}{L_0} = \frac{3.5}{1800} \\ &= 1.9444 \times 10^{-3} \text{ mm/mm}\end{aligned}$$

The Yield strength corresponds to the maximum elastic strain in the rod

$$\begin{aligned}\epsilon_Y &= \frac{\sigma_Y}{E} = \frac{250 \times 10^6}{200 \times 10^9} \\ &= 1.25 \times 10^{-3} \text{ mm/mm}\end{aligned}$$

Since $\epsilon_{\text{tot}} > \epsilon_Y$, the rod goes into plastic deformation range.



IN-CLASS EXAMPLE (cont'd)

Solutions, cont'd

According to the geometry (see the figure), we know that, after the loading is removed, the recovered strain (elastic part of deformation) is exactly equal to ε_Y , i.e. $\varepsilon_{el} = \varepsilon_Y$.

Since $\varepsilon_{el} + \varepsilon_{pl} = \varepsilon_{tot}$, we have:

$$\varepsilon_{pl} = \varepsilon_{tot} - \varepsilon_{el} = 1.9444 \times 10^{-3} - 1.25 \times 10^{-3} = 0.69444 \times 10^{-3} \text{ mm/mm}$$

The permanent set (δ_p) can be obtained by definition of plastic strain:

$$\delta_p = \varepsilon_{pl} \times L_0 = 0.69444 \times 10^{-3} \times 1800 = 1.25 \text{ mm} \quad \textbf{(Ans.)}$$

Since the rod goes into plastic deformation, the maximum value of the force P should induce internal stress equal to the Yield strength:

$$P = \sigma_Y \times A_0 = \sigma_Y \times \frac{\pi}{4} D^2 = 250 \times 10^6 \times \frac{\pi}{4} \times 0.032^2 = 201.06 \text{ kN} \quad \textbf{(Ans.)}$$